Deep Learning Diagnostic Tools for Digital Chemical Pathology

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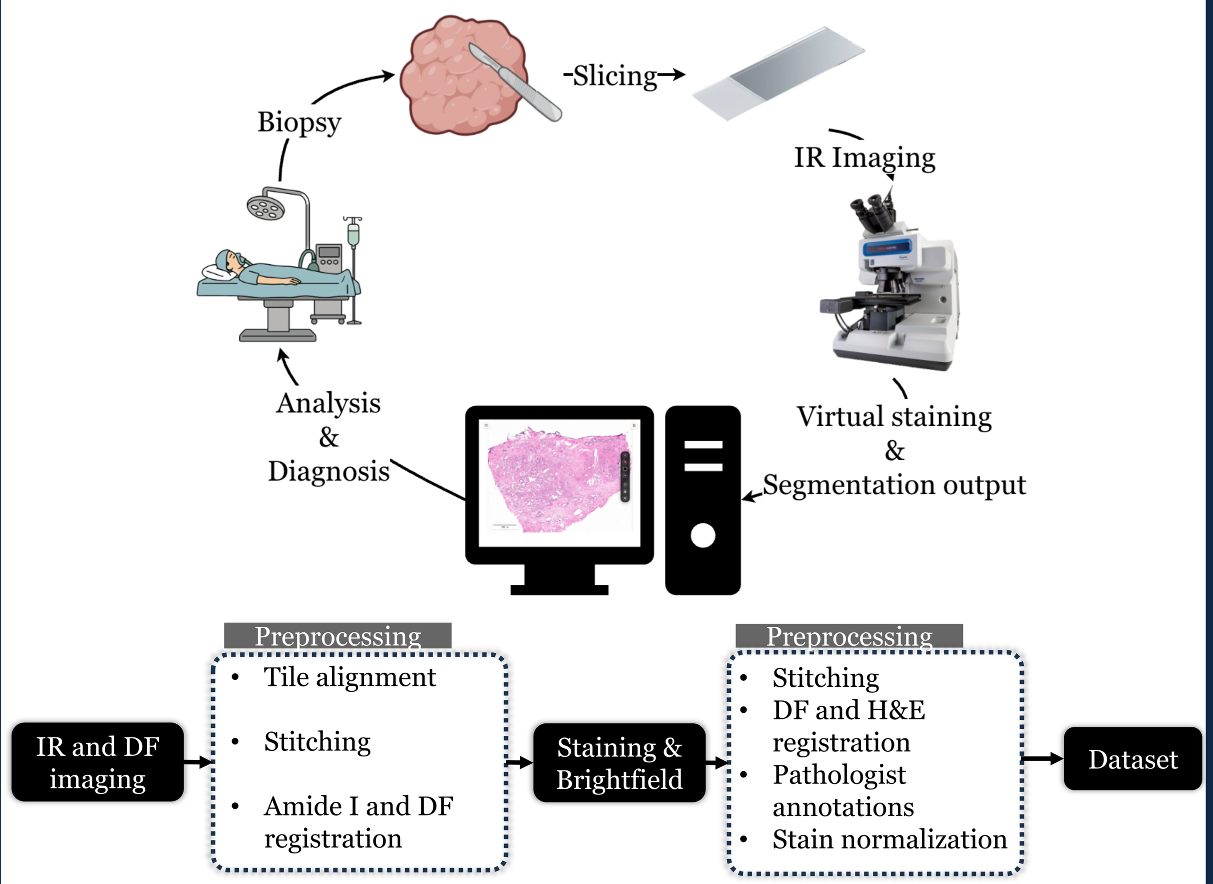
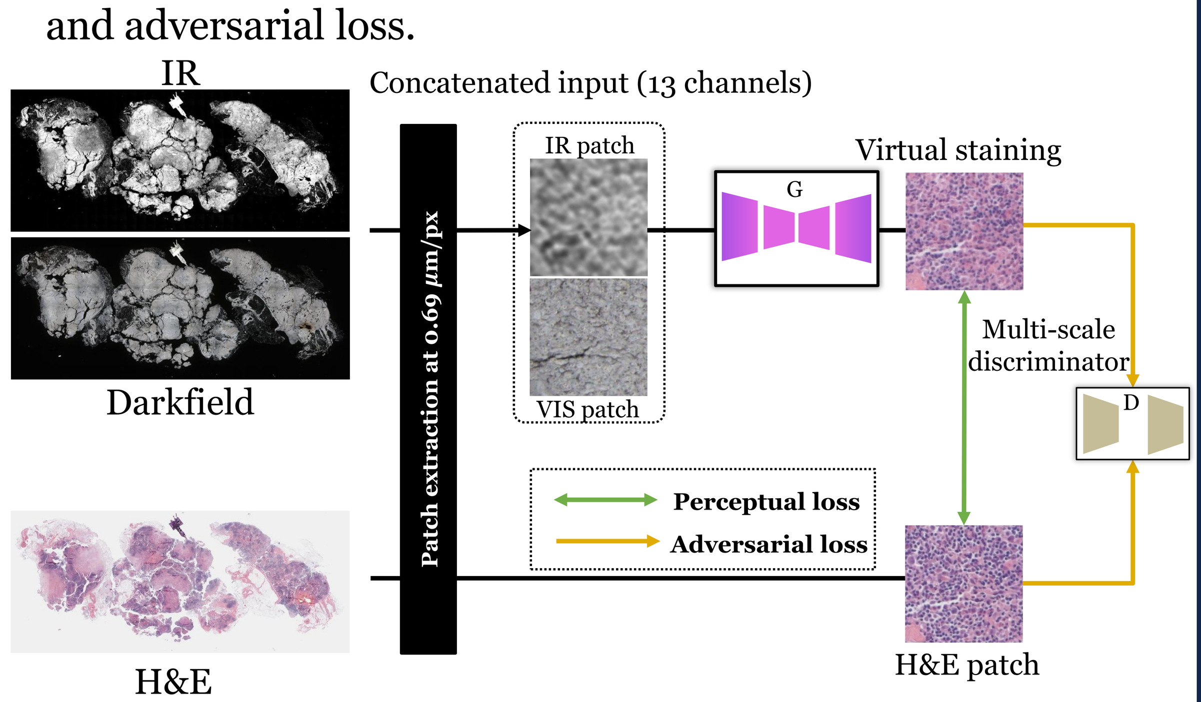
Motivation

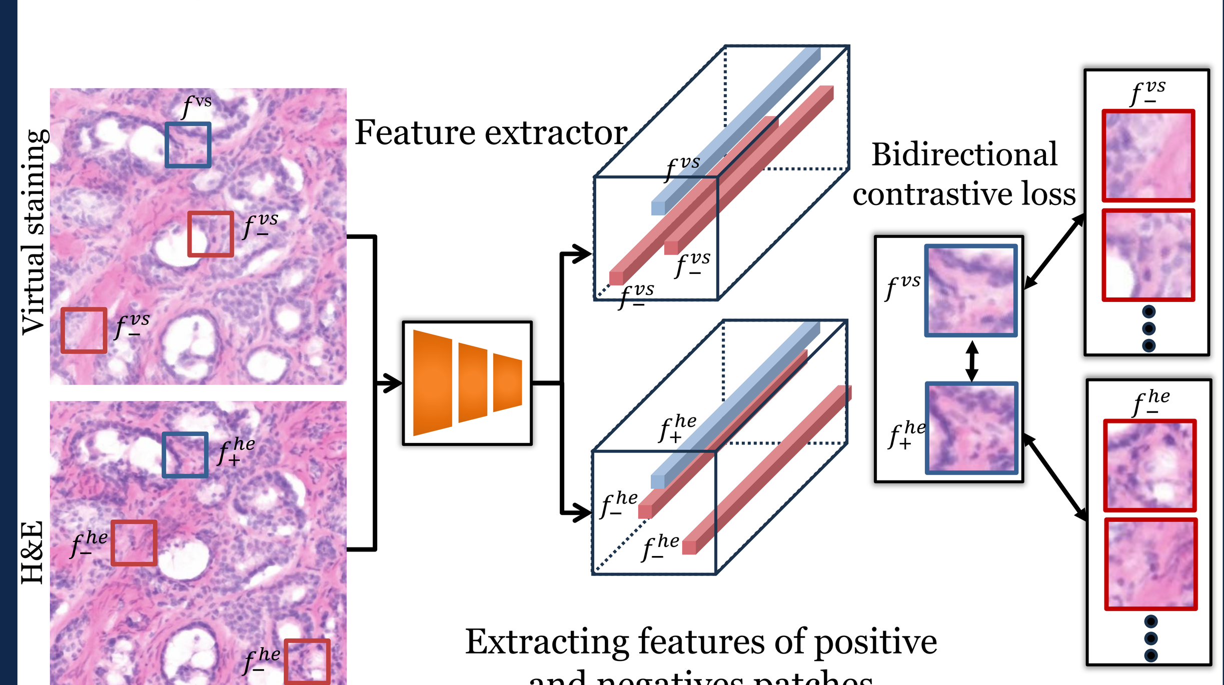
Conventional pathology is labor-intensive and time-consuming, involving tissue preparation, staining of biopsy material, and manual microscopic review, often stretching turnaround to several days between the initial biopsy and the diagnostic report. Deep learning (DL) algorithms can use the chemical contrast provided by mid-infrared absorption to identify cancer pathologies and transform infrared (IR) images into interpretable stains for fast, label-free pathology [1-3].



Virtual Staining

Virtual staining is performed to transform darkfield (DF) and IR images into synthetic hematoxylin and eosin (H&E) images. This is done using a ResNet50-based U-Net Generator with CBAM attention trained with perceptual and adversarial loss.



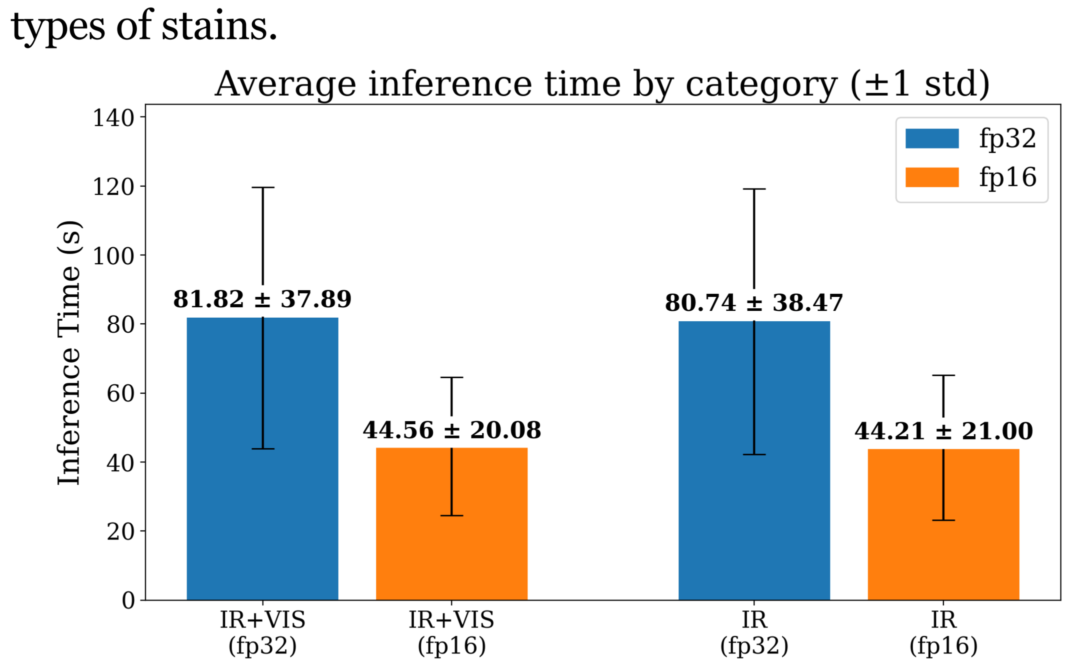


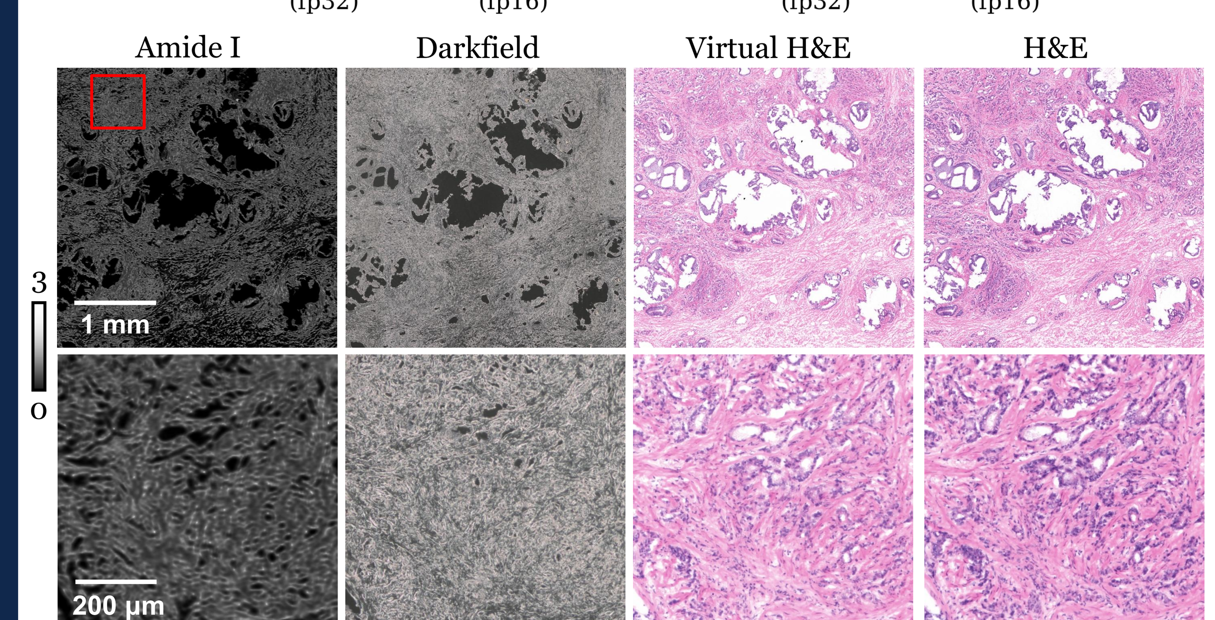
Speed Optimizations

For IR-based virtual staining to enhance traditional histopathology, it must be accurate, rapid, and diagnostically actionable.

In order to reduce inference time, we used TensorRT to quantize our model from floating point 32 (fp32) to floating point 16 (fp16), resulting in a 45.53% and 45.25% increase in inference speed on average across 239 samples. Inference was done on a Dell Precision 7820 workstation with an Intel Xeon W-2265 and Nvidia A4500 GPU.

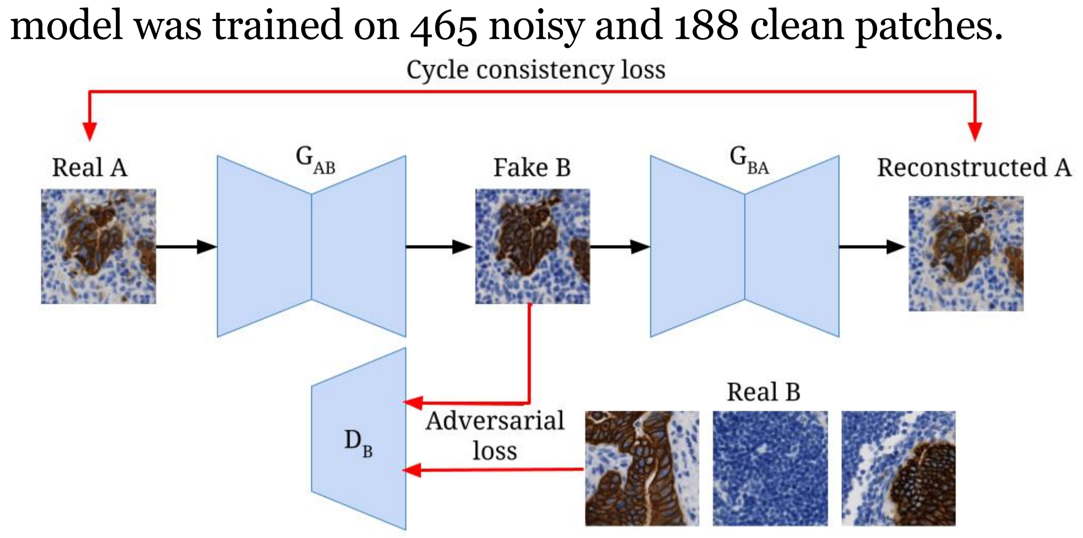
In the future, we hope to expand virtual staining to other types of stains.





Cytokeratin Stain Correction

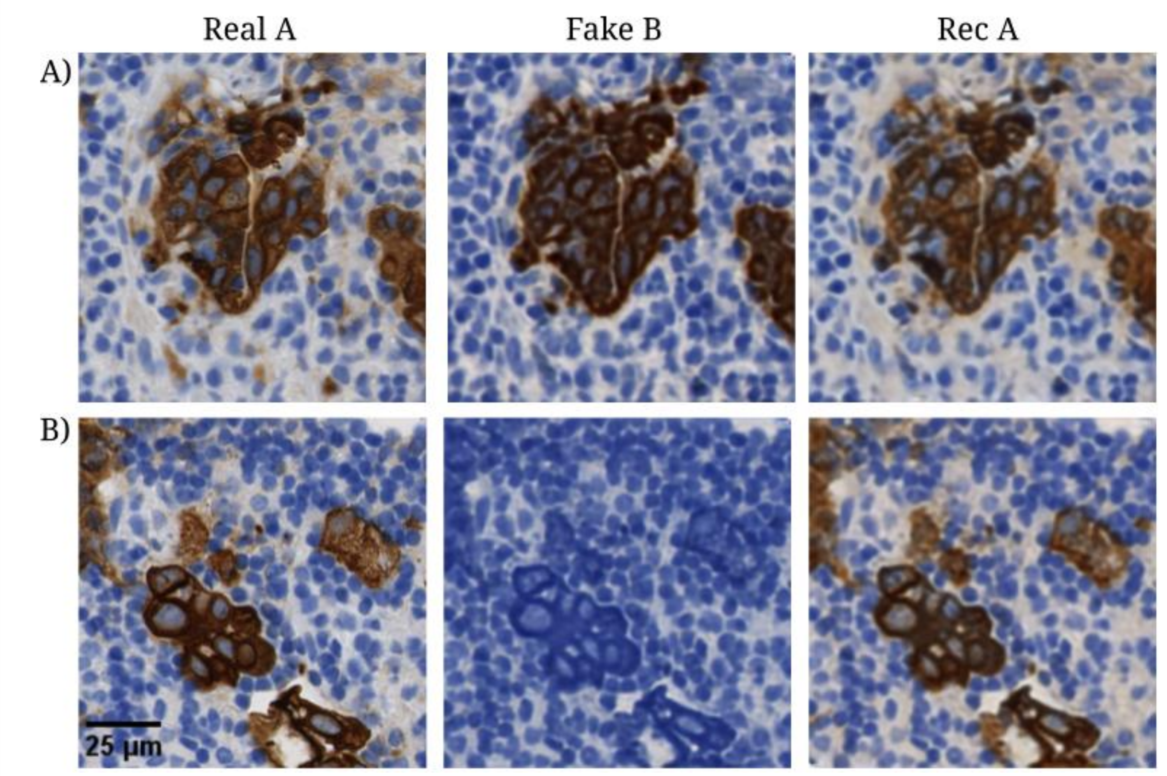
Non-specific binding of cytokeratin (CK) results in noisy ground truth immunohistochemistry (IHC) images that interfere with the virtual staining process. A data cleaning pipeline using cycleGAN was used to synthesize 'clean' IHC images. IHC images were divided into 256x256 pixel patches and classified as noisy and clean. The cycleGAN model was trained on 465 noisy and 188 clean patches.



Adapted from [5]

Results and Discussion

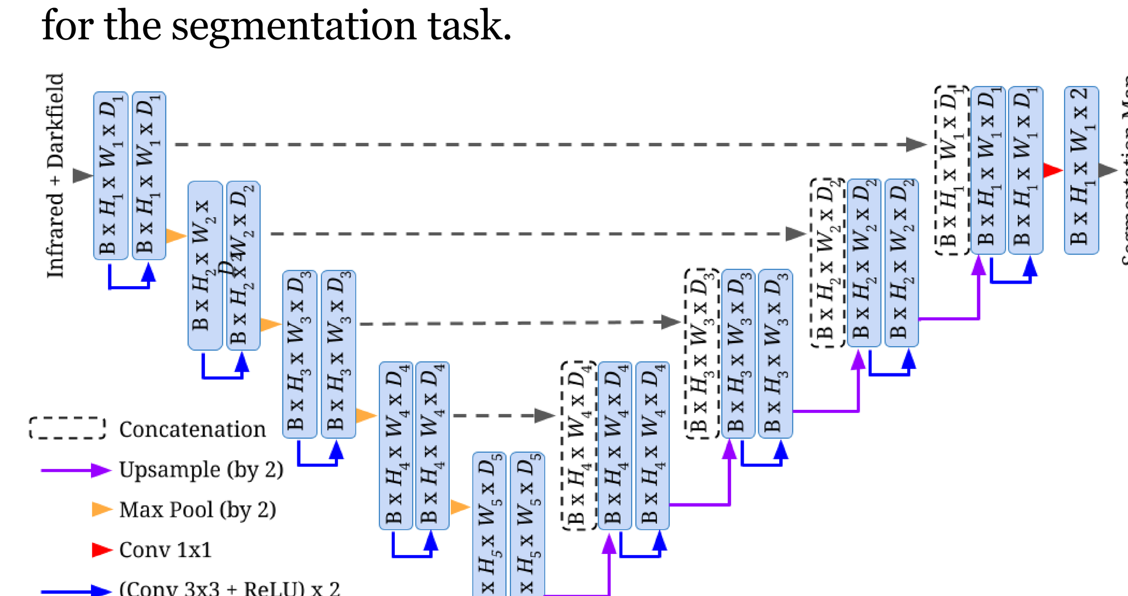
While the cycleGAN preserves the cancer stain for larger areas, it tends to overestimate the cancer region and also miss smaller regions. The following figure shows a successful (A) and a failure (B) case of stain correction.



Automated Cytokeratin Detection

Dataset and Model Architecture

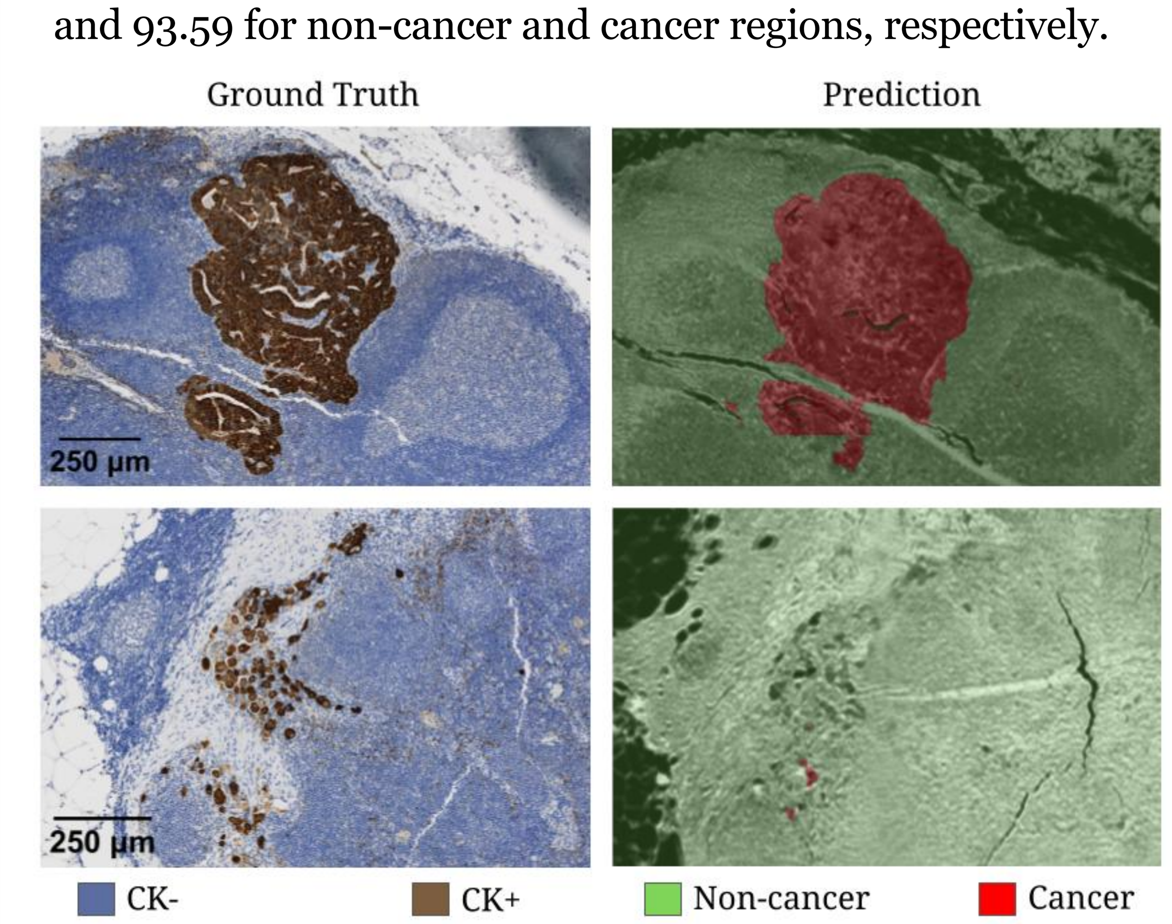
IR and DF images of endometrial lymph node tissue biopsies were model inputs. CK IHC slides from parallel sections were used for precise metastatic (CK+) vs normal (CK–) annotations. 22 whole-slide images (WSIs) were used for training/validation, and 3 WSIs were held out for testing. Images were downsampled to 4.57 µm/pixel, normalized to the dataset mean and standard deviation. An Attention U-Net model [4] with cross-entropy loss was used for the segmentation task.



Adapted from [4]

Results and Discussion

The Attention U-Net model can reasonably detect large areas of cancer, but has trouble detecting smaller areas of cancer. The model had a validation F1-score (%) of 99.97 and 93.59 for non-cancer and cancer regions, respectively.

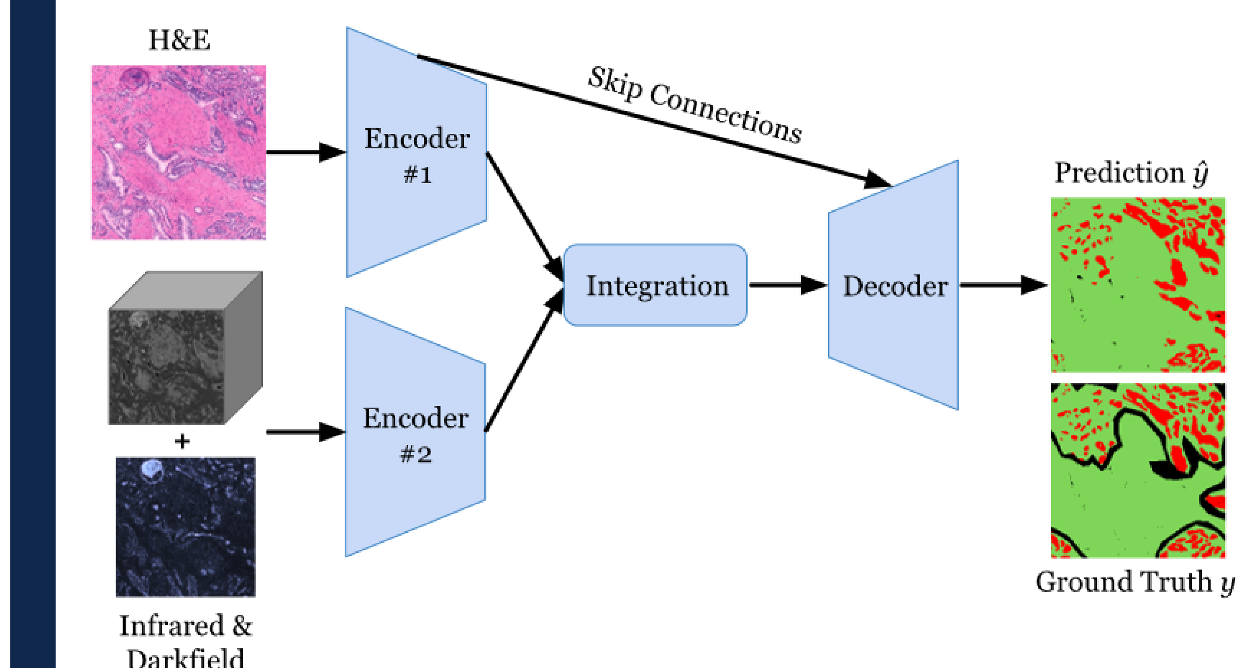


Tri-Modality Model for Prostate Cancer Detection

Dataset & Architecture

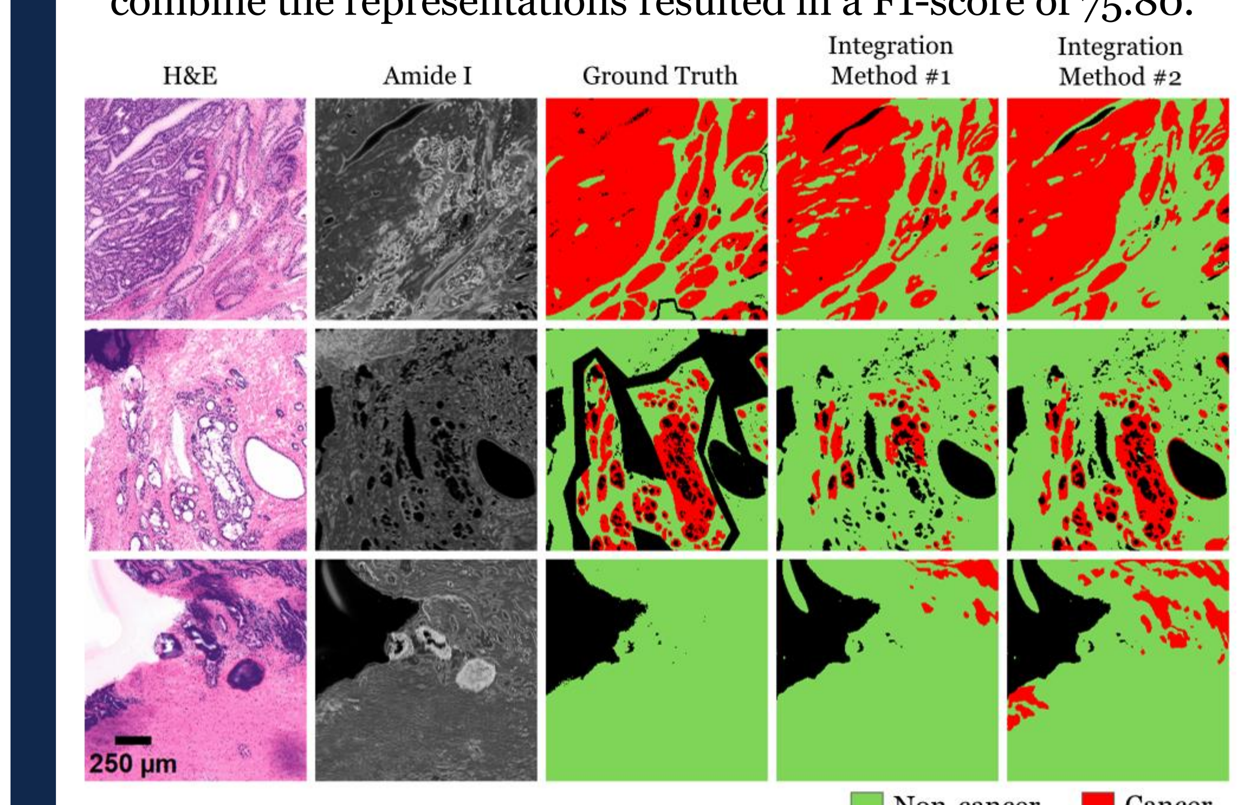
51 prostate biopsies were downsampled to 6 µm/pixel, normalized to the training dataset mean and standard deviation, and augmented with random flips and rotations.

A CNN consisting of two H&E and IR + darkfield encoders were created to leverage all three imaging modalities. Concatenation and ScConv [6] were used to combine representations before decoding into a segmentation map.



Results

The concatenation approach (Method #1) achieved a F1-score (%) of 71.68. The use of ScConv (Method #2) to combine the representations resulted in a F1-score of 75.80.



Discussion

Multi-modality models have the potential to improve cancer detection. Future research should focus on strategic modality integration techniques.

Acknowledgements & References

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[1] Yeh et al., Infrared spectroscopic laser scanning confocal microscopy for whole-slide chemical imaging, *Nature Communications* (2023).
[2] Falahkheirkhah et al., Accelerating cancer histopathology with chemical imaging and machine learning, *Cancer Research Communications* (2023).
[3] Falahkheirkhah et al., Deep learning-based protocols to enhance infrared imaging systems, *Chemometrics Intelligent Laboratory Systems* (2021).
[4] Oktay et al., Attention U-Net: Learning where to look for the pancreas, *MIDL* (2022).
[5] Tu et al., Multi-style image transfer system using conditional cycleGAN, *The Imaging Science Journal* (2021).
[6] Li et al., SCConv: Spatial and channel reconstruction convolution for feature redundancy, *CVPR* (2023).
[7] Andonian et al., Contrastive feature loss for image prediction, *ICCVW* (2021).
[8] Liu et al., Understanding reference policies in direct preference optimization, *arXiv* (2024).